



Effects of Computed Tomography Technical Parameters on Body-Composition Analysis

Jin Young Yoo, Moon Hyung Choi

Department of Radiology, Eunpyeong St. Mary's Hospital, College of Medicine, The Catholic University of Korea, Seoul, Republic of Korea

Body-composition analysis (BCA) is gaining increasing clinical importance, because abnormalities in muscle and fat distribution are closely associated with patient outcomes for various diseases. Although several methods for assessing body composition are available, including bioelectrical impedance analysis, dual-energy X-ray absorptiometry, and magnetic resonance imaging, computed tomography (CT) has emerged as the most widely used imaging modality owing to its accuracy, accessibility, and artificial intelligence-driven automated analytical capabilities. CT-based BCA enables the precise quantification of skeletal muscle and adipose tissues, but its measurements can be influenced by various technical factors, such as the contrast phase, tube current and voltage, slice thickness, reconstruction algorithm, and scanner type. These parameters particularly affect attenuation-based metrics such as muscle density. Recent technological advancements, such as iterative reconstruction, dual-energy CT, and photon-counting CT, have resulted in new capabilities but may further introduce variability. This review summarizes the effects of CT parameters on BCA results and underscores the need for awareness and consistency when performing CT-based BCA. A better understanding of these factors may improve measurement reproducibility and support broader clinical and research applications.

Keywords: Body-composition analysis; CT acquisition parameters; CT reconstruction techniques; Sarcopenia; Obesity

INTRODUCTION

Body-Composition Analysis

Body-composition analysis (BCA), which quantifies fat and muscle compartments, is increasingly used to assess metabolic health and physical function as well as disease prognosis in clinical practice. Among the abnormalities revealed by BCA, sarcopenia and obesity represent two clinically important states: i.e., muscle deficiency and excess fat, respectively. Sarcopenia, which was originally defined as age-related muscle loss, is now considered a complex disease characterized by reduced strength and

low muscle quantity or quality [1,2] and is associated with increased risks of falls, disability, hospitalization, and mortality. Obesity, typically defined by excessive body fat and commonly evaluated by using the body mass index (BMI ≥ 30), is also linked to adverse health outcomes. However, BMI cannot be used to distinguish between lean mass and fat mass and may misclassify individuals with atypical body composition [3]. The coexistence of these two conditions gives rise to sarcopenic obesity, a phenotype characterized by simultaneous muscle loss and fat accumulation. This condition, which is particularly prevalent in older adults, is associated with poorer clinical outcomes than either sarcopenia or obesity alone [1,3,4].

Various methods are available to assess body composition, including bioelectrical impedance analysis (BIA), dual-energy X-ray absorptiometry (DEXA), and imaging-based techniques. BIA is convenient and inexpensive; however, its accuracy is affected by hydration, food intake, and physical activity, and it cannot be used to precisely differentiate fat compartments. DEXA offers regional estimates of bone mass, lean mass, and fat mass but has limitations in assessing visceral and intramuscular fat, particularly in

Received: August 5, 2025 **Revised:** October 20, 2025

Accepted: October 20, 2025

Corresponding author: Moon Hyung Choi, MD, PhD, Department of Radiology, Eunpyeong St. Mary's Hospital, College of Medicine, The Catholic University of Korea, 1021 Tongil-ro, Eunpyeong-gu, Seoul 03312, Republic of Korea

• E-mail: cmh@catholic.ac.kr

This is an Open Access article distributed under the terms of the Creative Commons Attribution Non-Commercial License (<https://creativecommons.org/licenses/by-nc/4.0>) which permits unrestricted non-commercial use, distribution, and reproduction in any medium, provided the original work is properly cited.

obese or fluid-overloaded patients [5,6]. Cross-sectional imaging modalities such as computed tomography (CT) and magnetic resonance imaging (MRI) are considered the reference standards for BCA. These techniques offer detailed anatomical visualization and allow the quantification of both fat and muscle tissue. CT is valuable because of its widespread use in routine clinical care and its ability to facilitate opportunistic BCA based on existing scans. MRI offers comparable or even superior accuracy in muscle assessment without radiation exposure and enables the direct evaluation of both muscle volume and fat infiltration using specific sequences. Ultrasound is portable and inexpensive but has limited reproducibility and lacks standardized cutoff values, limiting its broader clinical application [1,5-8].

CT-Based BCA

The clinical relevance of CT-based BCA is well-established across a range of clinical contexts [9-11]. Recent advances in deep-learning algorithms have enabled the automated segmentation of abdominal muscles and adipose tissue, thereby accelerating research throughput and expanding large-scale cohort studies [12]. CT-based analysis quantifies skeletal muscle and adipose tissues based on differences in tissue radiodensity, which is expressed in Hounsfield units (HU) [13]. Cross-sectional images at the third lumbar vertebral level (L3) are most commonly used for analysis, given their strong correlation with whole-body muscle and fat volumes [14,15].

The skeletal muscle area (SMA) is typically segmented using attenuation thresholds of -29 to +150 HU, which were originally established on 120-kV CT images. The measured SMA is normalized to the patient height squared to calculate the skeletal muscle index (SMI, cm^2/m^2), which is widely used as a surrogate for diagnosing sarcopenia [7,14,16]. Advanced classifications of the SMA, such as the normal-attenuation muscle area (NAMA; +30 to +150 HU) and low-attenuation muscle area (LAMA; -29 to +29 HU), which is also referred to as the steatotic muscle area, enable a more detailed characterization of muscle integrity and fat infiltration [17]. Muscle quality is assessed by the mean attenuation of muscle tissue: lower values indicate fat infiltration, which is known as myosteatosis [6,14,17]. Intermuscular adipose tissue (IMAT), which is defined as the fat interspersed between or within muscle groups or fascicles, is quantified using attenuation thresholds that range from -190 to -30 HU and serves as another indicator

of myosteatosis [17]. Adipose tissue is typically segmented into visceral adipose tissue (VAT) and subcutaneous adipose tissue (SAT), primarily on the basis of anatomical location, although HU thresholds (approximately -190 to -30 HU) are applied.

Heterogeneity in CT acquisition and reconstruction has emerged as a major source of variability in BCA [18]. Because key BCA metrics are HU threshold-based, changes in acquisition and reconstruction can alter HU features; consequently, the metrics themselves may vary. Muscle attenuation values, which indicate muscle quality and fat infiltration, are particularly sensitive to changes in CT parameters [19]. Nevertheless, many studies have failed to report basic technical details such as scan indications, anatomical coverage, contrast use/phase, and other CT parameters, which limits the reproducibility of their findings [14]. Although CT-based BCA demonstrates clinical utility even without standardized protocols, variability introduces the risk of inconsistency, especially in longitudinal or multicenter studies.

Accordingly, this review provides a comprehensive overview of how CT technical parameters, including acquisition and reconstruction protocols, and scanner-specific characteristics, affect BCA (Fig. 1). Special emphasis was placed on abdominal CT, which is the most widely used modality for CT-based BCA in clinical practice.

VARIABLES RELATED TO CT ACQUISITION

Contrast Material

An iodinated contrast medium is commonly used in abdominal CT to improve the visualization of soft-tissue structures with similar baseline attenuation. Although the contrast enhancement of skeletal muscle and adipose tissue is generally less pronounced than that of intra-abdominal organs, it can still affect the BCA, particularly in attenuation-based measurements, such as NAMA and LAMA within the SMA. Figure 2 illustrates the differences in BCA results between pre- and post-contrast images, accompanied by HU histograms in skeletal muscle pixels.

A scoping review identified 10 studies that evaluated the effects of contrast agents on muscle attenuation, seven of which reported significant increases following contrast agent administration [18]. The SMA was also reported to be influenced by contrast agents in seven studies, although the direction and magnitude of changes were inconsistent and typically small. Nonetheless, post-contrast images generally

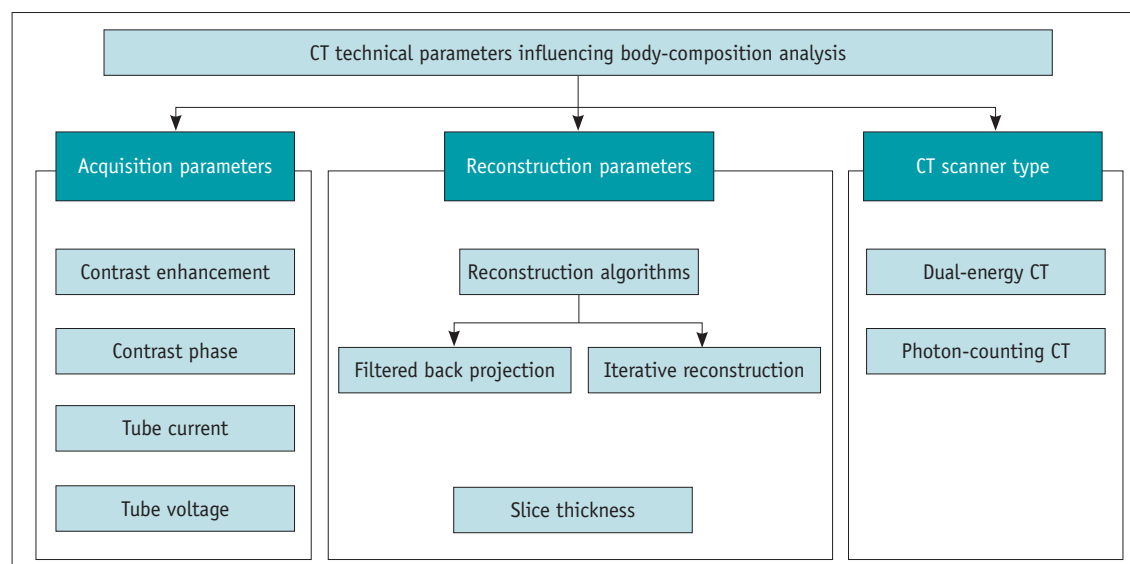


Fig. 1. CT technical parameters that potentially affect body-composition analysis. CT = computed tomography

demonstrate greater SMA and skeletal muscle density (SMD) regardless of the CT acquisition or injection protocol used [20-22]. For adipose tissue, one study reported a 7.6% average decrease in the VAT area and a 5.4% increase in VAT attenuation for post-contrast images, which likely reflects the enhancement of fat adjacent to vessels and organs [20]. In contrast, the SAT and SMA increased only marginally (0.1% and 0.2%, respectively).

Despite the good-to-excellent positive linear correlation between the pre- and post-contrast values of the SMA and SMD, the two are not directly interchangeable [20,23]. For example, pre-contrast values were approximated by applying a constant correction of -7.5 HU to post-contrast muscle attenuation values, but such corrections are not universally valid [20]. Accordingly, mixing pre- and post-contrast scans is generally not recommended. In retrospective settings or when CT examinations acquired for different purposes must be combined, presenting variability analyses and applying corrections or normalization when feasible may improve comparability. For longitudinal assessments, the consistent use of a single phase is essential to avoid parameter-related bias.

Contrast enhancement is influenced by multiple factors, including scan parameters, patient physiology, and injection protocol. Injection-related variables such as contrast medium type, iodine concentration, injection rate, and volume can significantly alter tissue enhancement [24-26]. Moreover, the patient's body composition can modulate contrast enhancement, leading to inter-subject variability, even under the same injection settings [27].

While low-iodine-concentration contrast agents (e.g., 240 or 270 mg I/mL) and reduced volumes are increasingly being used to minimize renal toxicity, their effects on BCA have rarely been investigated [28-30]. Ethical constraints on repeated scans in the same patient have likely contributed to this limited evidence. Further research is warranted to clarify how contrast-protocol variations affect CT-based BCA.

Phase of CT Images

Because contrast enhancement on CT can affect muscle and fat attenuation, non-contrast images may provide more consistent values, particularly for attenuation-based metrics [31]. However, the routine acquisition of non-contrast images solely for the BCA is not feasible, and contrast-enhanced CT protocols are essential for abdominal imaging. These protocols often include single- or multiphase (e.g., arterial, portal venous, and delayed phases) studies, usually employing bolus-tracking techniques for optimized timing. Accordingly, multiple studies have investigated whether the CT phase influences the BCA.

One study found no significant difference in SMI, whereas muscle attenuation progressively increased across the non-contrast, arterial, and portal venous phases [32]. Several studies on three- or four-phase CT protocols have likewise reported gradual increases in the SMD and SMA from the unenhanced phase to the delayed phase [33-35]. However, for sarcopenia, the findings are inconsistent; one study reported a reduced prevalence of sarcopenia in the delayed phase [33], whereas others found no significant phase-

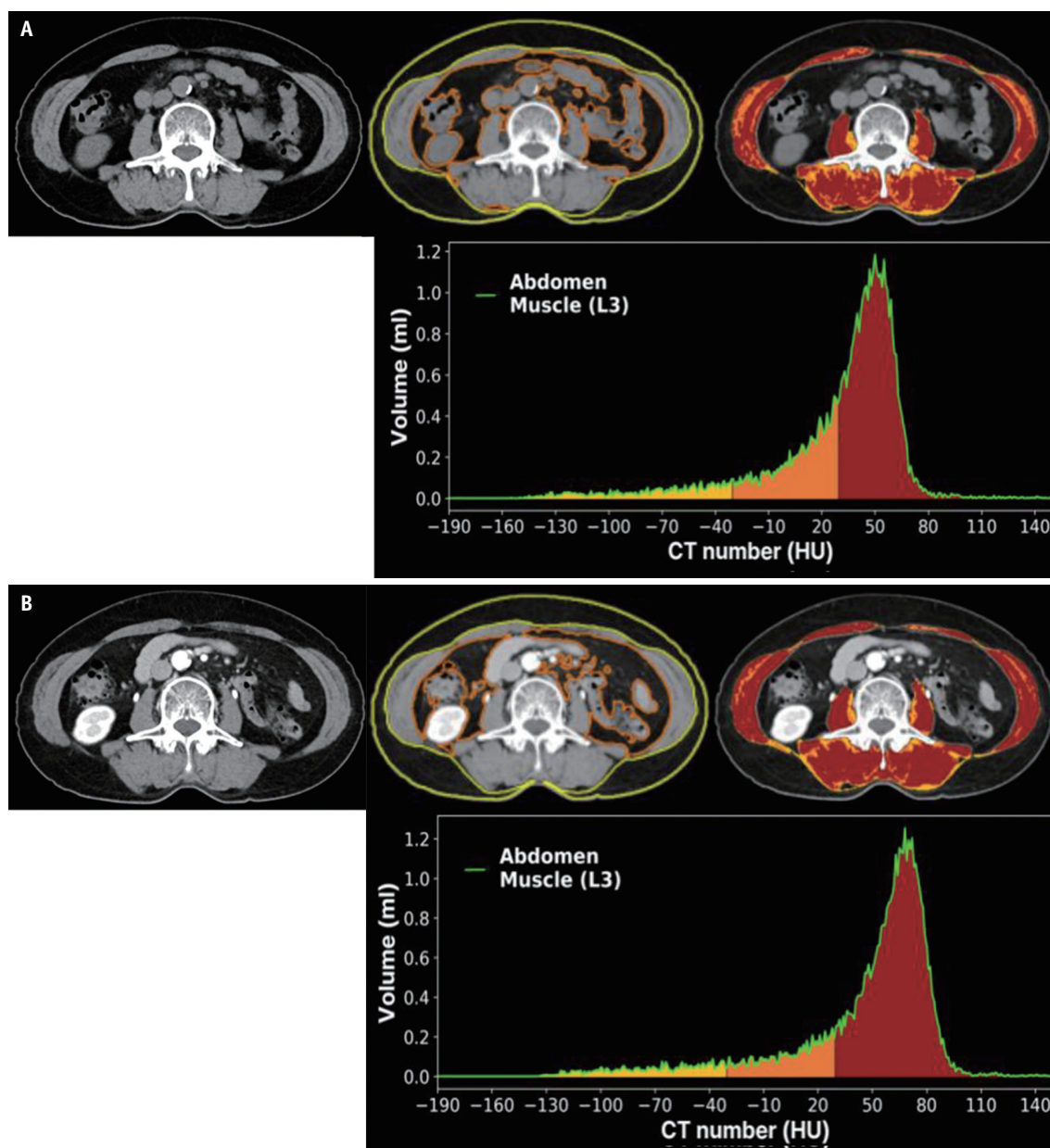


Fig. 2. Pre-contrast and post-contrast abdominal CT in a 69-year-old woman, acquired at 80 kV. **A, B:** The figure shows BCA results at the L3 level from CT images acquired pre- (**A**) and post-contrast (**B**). In each panel, the top row displays the original CT image (left), adipose tissue boundaries (middle; SAT = yellow outline, VAT = orange outline), and muscle segmentation (right; NAMA = red, LAMA = orange, IMAT area = yellow). The bottom row shows skeletal-muscle attenuation histograms. BCA metrics are reported as pre- vs. post-contrast image, with areas in cm² and SMD in HU: VAT area 88 vs. 77, SAT area 115 vs. 113, SMA 90 vs. 93, SMD 39.2 ± 23.0 vs. 55.4 ± 26.0 , IMAT area 8 vs. 7, LAMA 24 vs. 14, and NAMA 65 vs. 80. The histograms demonstrate a rightward shift in the skeletal-muscle HU distribution after contrast administration. Automated analyses were performed using ClariMetabo version 1.1.0 (ClariPi, Seoul, Korea). CT = computed tomography, BCA = body-composition analysis, SAT = subcutaneous adipose tissue, VAT = visceral adipose tissue, NAMA = normal-attenuation muscle area, LAMA = low-attenuation muscle area, IMAT = intermuscular adipose tissue, SMD = skeletal muscle density, HU = Hounsfield units, SMA = skeletal muscle area

related differences [34,35]. A more detailed analysis showed that SMA, NAMA, and SMD increased in later phases, whereas LAMA decreased, leading to fewer patients being classified as having myosteatosis [36]. The VAT

area demonstrated strong correlations across phases but gradually decreased with delayed imaging [33,35]. When predefined thresholds were applied, the prevalence of visceral obesity significantly decreased in the later phases

[33]. A perfusion CT study also showed a time-dependent increase in SMI following contrast agent injection, with reductions in LAMA and the adipose tissue index, defined as

the adipose tissue area normalized by height squared [37].

In summary, although SMIs and sarcopenia classifications remained consistent across the phases, SMA and SMD

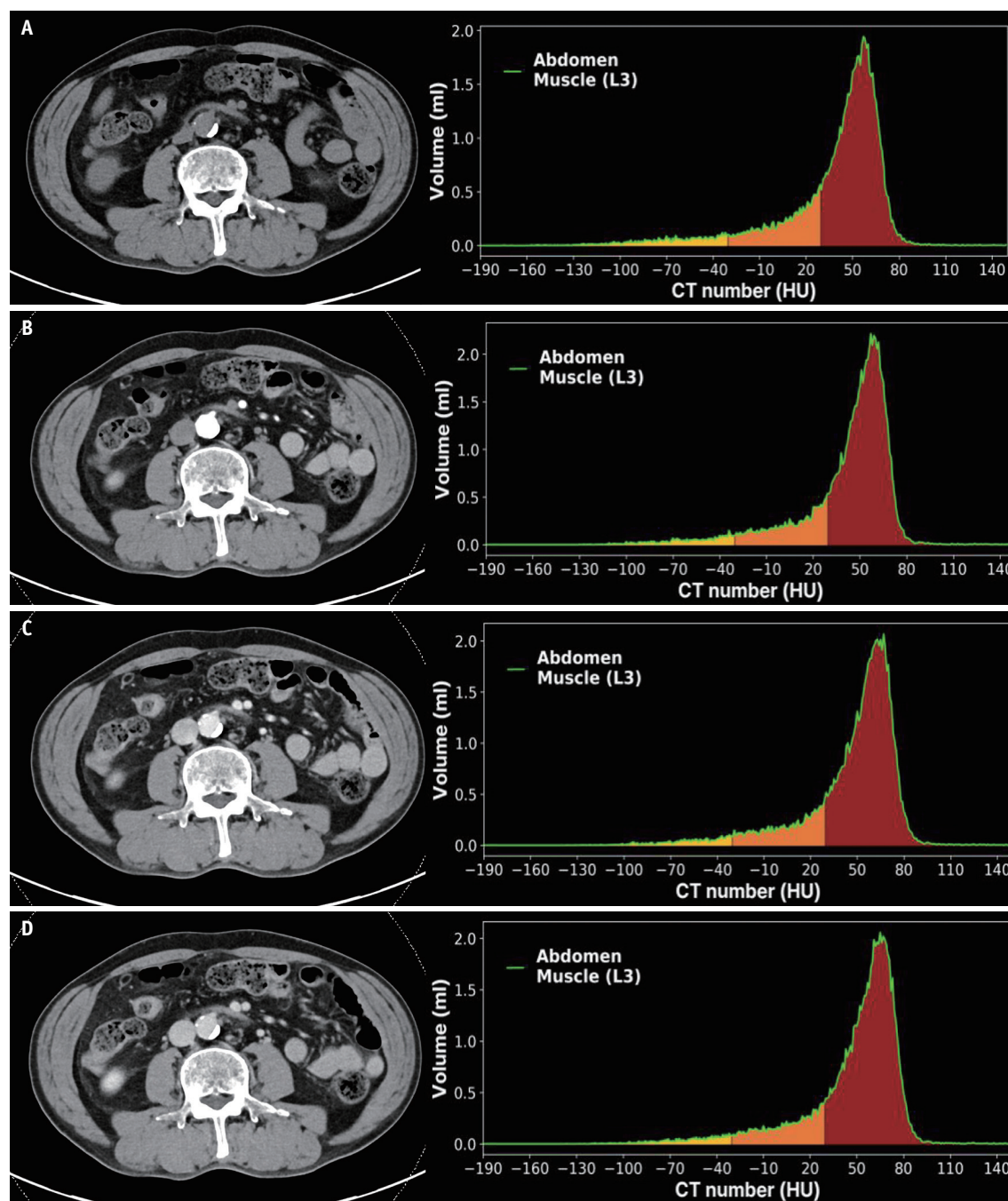


Fig. 3. Liver dynamic CT with four phases (pre-contrast, arterial, portal venous and delayed) in a 65-year-old man. **A-D:** The figure shows the BCA results at the L3 level from dynamic CT images acquired in the pre-contrast (**A**), arterial (**B**), portal venous (**C**), and delayed (**D**) phases. Each panel displays the original CT image (left) and skeletal muscle attenuation histograms (right, NAMA = red; LAMA = orange; IMAT area = yellow). As shown in the histograms, from the pre-contrast phase to the delayed phase, the number of pixels in the IMAT area decreased, whereas the number of pixels in the NAMA increased. BCA metrics (pre-contrast/arterial/portal venous/delayed phase; areas in cm², SMD in HU): VAT area (109/90/87/90), SAT area (45/42/41/41), SMA (136/148/148/149), SMD (44.6 ± 22/46.0 ± 23/50.4 ± 24/51.8 ± 24), IMAT area (9/7/6/6), LAMA (27/27/23/23), NAMA (109/121/125/127). CT = computed tomography, BCA = body-composition analysis, NAMA = normal-attenuation muscle area, LAMA = low-attenuation muscle area, IMAT = intermuscular adipose tissue, SMD = skeletal muscle density, HU = Hounsfield units, VAT = visceral adipose tissue, SAT = subcutaneous adipose tissue, SMA = skeletal muscle area

generally increased, whereas LAMA and adipose tissue area tended to decrease in the later phase. These findings suggest that abdominal muscle and fat tissues gradually enhance after contrast injection and that CT phase selection can influence BCA results. Figure 3 illustrates BCA results for a representative case across the four phases of dynamic liver CT.

Tube Current

Reducing the tube current is a widely adopted strategy for lowering the radiation dose in CT imaging, because radiation exposure is directly proportional to the tube current. A decrease in the tube current leads to a reduced number of X-ray photons generated per unit time, which results in increased image noise and decreased signal-to-noise ratio (SNR) [38]. Low-tube current CT protocols have been implemented for the abdomen, particularly in patients with suspected renal colic [39,40]. Figure 4 presents BCA results from CT images simultaneously acquired at two different tube currents on a dual-source CT system. Although fixed low-tube current protocols are not routinely used in abdominal CT, automatic tube current modulation

(ATCM) is common. With a fixed tube current, the image noise varies according to patient size and body part attenuation. In contrast, ATCM dynamically adjusts the tube current along the z-axis and x-y plane to maintain a consistent image quality throughout the scan range [38]. With ATCM, a reference value is predefined before scanning to represent the desired image quality, typically that of an 80-kg standard patient. This value is referred to as the effective mAs, reference case, or image quality level, depending on the vendor. The final reported tube current is expressed as the mean value across the scan range. Although ATCM effectively reduces radiation exposure and ensures uniform image quality, its direct effect on BCA remains insufficiently studied. Nonetheless, distinguishing between a fixed tube current and ATCM is essential for the accurate interpretation of scan parameters in CT-based BCA.

A phantom study revealed that reducing the tube current to 10%–50% of the standard dose yielded a similar SMA but lower muscle attenuation [41]. Another study comparing diagnostic CT with ATCM (reference mAs = 200) and low-dose PET/CT attenuation correction scans (a fixed tube

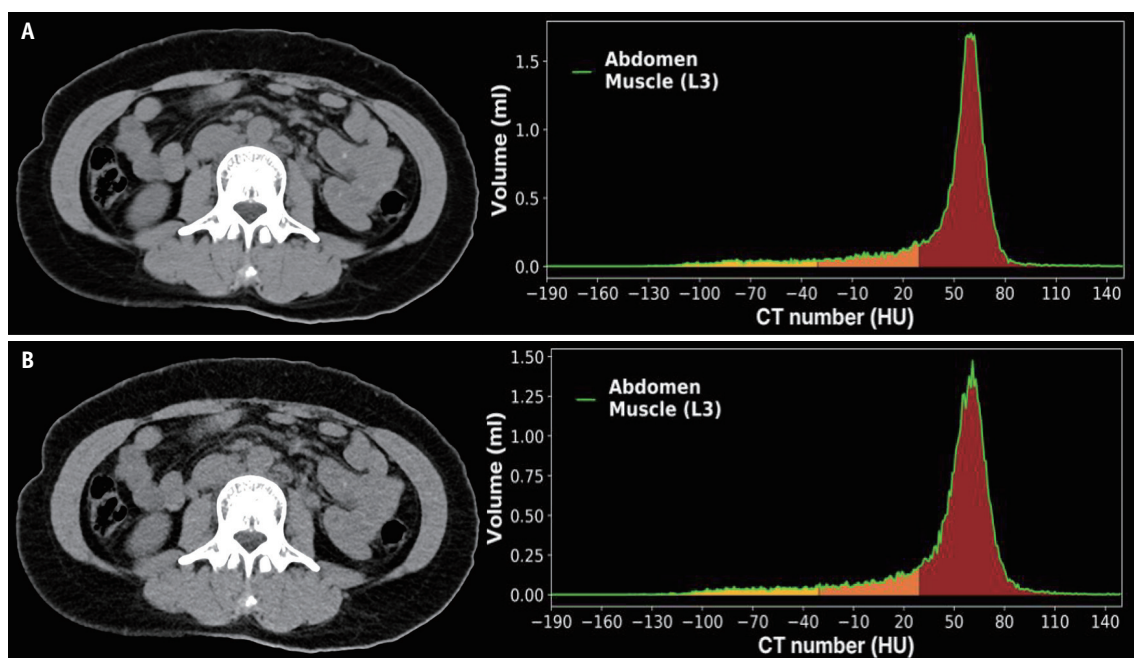


Fig. 4. Non-contrast abdominal CT at 90 kV in a 39-year-old woman, acquired on a dual-source scanner with two tube-current settings (240 mAs vs. 60 mAs). **A, B:** The figure shows BCA results at the L3 level from CT images acquired at 240 mAs (**A**) and 60 mAs (**B**). Each panel displays the original CT image (left) and skeletal muscle attenuation histograms (right, NAMA = red; LAMA = orange; IMAT area = yellow). As shown in the histograms, the 240-mAs image exhibits a narrower, higher peak than the 60-mAs image, reflecting the reduced noise at a higher tube current. BCA metrics are reported as 240 mAs vs. 60 mAs, with areas in cm² and SMD in HU: VAT 47 vs. 47, SAT 124 vs. 124, SMA 81 vs. 81, SMD 52.1 ± 21.0 vs. 52.3 ± 22.0, IMAT area 6 vs. 6, LAMA 10 vs. 10, and NAMA 71 vs. 71. CT = computed tomography, BCA = body-composition analysis, NAMA = normal-attenuation muscle area, LAMA = low-attenuation muscle area, IMAT = intermuscular adipose tissue, SMD = skeletal muscle density, HU = Hounsfield units, VAT = visceral adipose tissue, SAT = subcutaneous adipose tissue, SMA = skeletal muscle area

current of 33–120 mA based on BMI) reported a lower SMA but higher SMD in low-dose images [21]. Histogram analysis revealed broader pixel-value distributions in the low-dose scans, suggesting that increased noise is a possible source of variation.

In one study, CT images acquired with low- and standard-tube current–time products (mean 28.8 mAs vs. 161.9 mAs) had significantly lower VAT and SAT areas and attenuation in low-dose images [42]. Although another study reported excellent correlations between diagnostic and low-dose images for the SMA, VAT, and SAT, it did not directly compare absolute values [43]. In that study, fixed tube current and voltage (40–160 mAs and 120 kV, respectively) were used for low-dose scans, whereas variable tube current and voltage (65–389 mAs and 100–130 kV, respectively) were employed for diagnostic scans. The presumed use of ATCM may have minimized the actual radiation dose difference, potentially explaining the stronger correlations than those reported in other studies.

These findings suggest that reducing the tube current introduces variability in the BCA. However, increased noise

from a lower tube current has an inconsistent impact on the SMD across studies. Because both upward and downward shifts have been reported, the net bias remains uncertain and warrants cautious interpretation. The degree of current reduction also influences variability, but the minimum acceptable threshold for a reliable BCA does not have a consensus definition. Finally, the use of ATCM should be clearly described in BCA studies, because patient-specific modulation may affect attenuation-based measurements, even when the same reference mAs is applied.

Tube Voltage

The tube voltage determines the effective energy of the X-ray beam and has an exponential relationship with radiation dose. A higher tube voltage improves beam penetration but also increases radiation exposure. Although 120 kV has traditionally been the standard in CT imaging, lowering the voltage to 80 kV reduces the radiation dose and enhances iodine conspicuity by reducing the photon energy toward the K-edge of iodine [38]. This reduction also allows the use of a reduced volume or concentration of

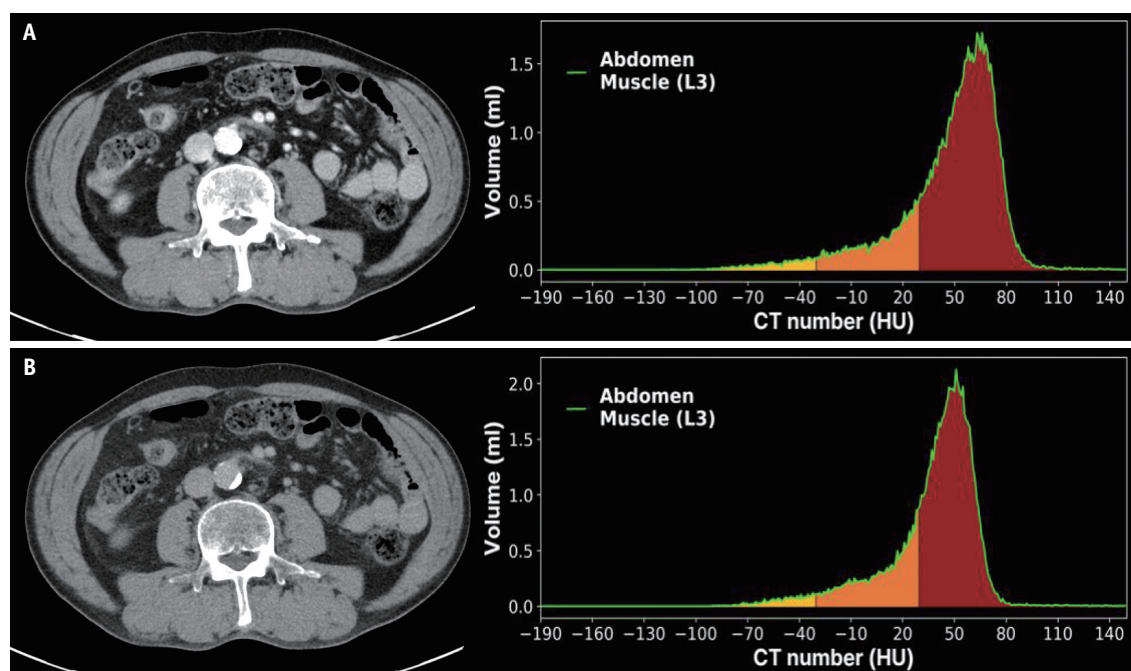


Fig. 5. Portal venous phase abdominal CT in a 65-year-old man, acquired on a dual-source dual-energy scanner with two tube-voltage settings (90 kV vs. 150 kV). **A, B:** The figure shows BCA results at the L3 level from CT images acquired at 90 kV (**A**) and 150 kV (**B**). Each panel displays the original CT image (left) and skeletal muscle attenuation histograms (right, NAMA = red; LAMA = orange; IMAT area = yellow). As shown in the histograms, the 90-kV images contain more pixels with higher HU values than the 150-kV images. BCA metrics are reported as 90 kV vs. 150 kV, with areas in cm² and SMD in HU: VAT area 85 vs. 86, SAT area 42 vs. 39, SMA 148 vs. 153, SMD 50.2 ± 25.0 vs. 40.2 ± 22.0, IMAT area 5 vs. 5, LAMA 26 vs. 37, and NAMA 122 vs. 116. CT = computed tomography, BCA = body-composition analysis, NAMA = normal-attenuation muscle area, LAMA = low-attenuation muscle area, IMAT = intermuscular adipose tissue, HU = Hounsfield units, SMD = skeletal muscle density, VAT = visceral adipose tissue, SAT = subcutaneous adipose tissue, SMA = skeletal muscle area

the contrast material without compromising the diagnostic quality [28,29]. Modern CT scanners often incorporate automatic tube-voltage selection based on a patient's attenuation profile from the scout view to maintain consistent image quality across body sizes. However, this introduces variability in the tube voltage, which can alter tissue attenuation values even without contrast material, highlighting the importance of protocol awareness in CT-based BCA. Figure 5 illustrates BCA results from two portal venous phase images acquired using dual-energy CT (DECT) at different tube voltage settings.

A scoping review reported that muscle attenuation increased at lower voltages across several studies, despite variations in measurement methods and voltage comparisons (e.g., 120 kV vs. 100 kV and 140 kV vs. 80 kV) [18]. In a phantom study comparing 120 kV and 80 kV, both SMA and SMD were greater at 80 kV, while NAMA decreased, and LAMA and IMAT increased [41]. In a clinical study using dual-source DECT (80 kV vs. 140 kV in the portal-venous phase), SMD was significantly greater, but both SMA and LAMA were significantly lower at 80 kV [44]. The discrepancy between the phantom and the clinical findings may have been due to the presence of contrast material in the clinical setting, because tissue attenuation is more sensitive to voltage changes in contrast-enhanced studies.

In the same study, fat attenuation was significantly greater in 140 kV images; however, the SAT and VAT indices (area/height²) did not differ significantly [44]. Another DECT study (100 kV vs. 150 kV in the portal venous phase) found that the SAT, VAT, and IMAT areas decreased with higher voltages, while attenuation values increased [45]. The observed increase in fat attenuation at higher voltages in both studies may reflect the combined effect of tube voltage and the minimal presence of contrast material within the fat compartments. Given these findings, simple area-based metrics and height-adjusted indices may yield different interpretations. Further research is warranted to clarify specific effects of tube voltage on fat quantification in CT-based BCA.

CT PARAMETERS RELATED TO RECONSTRUCTION

Reconstruction Algorithms

The CT image quality, particularly the texture and noise, depends on the reconstruction algorithm. During CT acquisition, X-rays are attenuated by internal tissues, recorded as projections from multiple angles, organized into

sinograms, and reconstructed into cross-sectional images using mathematical algorithms [46]. The conventional filtered back projection (FBP) method preprocesses raw projection data using filters (kernels) that balance sharpness and noise; soft kernels reduce noise but blur details, whereas sharp kernels enhance edge definition at the cost of more noise. A key limitation of FBP is that image noise increases at lower radiation doses, which degrades the image quality.

Iterative reconstruction (IR), enabled by improved computing power, simulates image formation and refines results iteratively, reducing noise and preserving image quality even at lower doses [38,46,47]. IR is available in various forms, including statistical IR and model-based IR, and typically offers user-defined strength levels (e.g., Levels 1–7). A known drawback of IR is the smoothing effect on image texture, which may affect radiodensity-based analyses such as CT-based BCA. The magnitude of this smoothing effect depends on both the reconstruction strength and the radiation dose [48].

A phantom study simulating vertebrae, abdominal muscle, and SAT at the lumbar level found no difference in SMA between FBP and IR but noted a small, consistent decrease in SMD (–0.1 to –1.2 HU) [41]. IR-reconstructed images also showed lower background noise and higher SNRs than image with FBP, suggesting that the reconstruction method may influence attenuation-based metrics, particularly for assessing small changes or performing longitudinal comparisons.

Most clinical studies on reconstruction methods have compared standard-dose FBP with low-dose IR, usually measuring tissue attenuation in regions of interest (ROIs) in organs rather than using full SMA segmentation. In a study comparing scans reconstructed with FBP and model-based IR at different time points and using different scanners, the radiation dose was lower with IR, psoas-muscle attenuation was unchanged, and perinephric-fat attenuation was lower [49]. Image noise was reduced in both muscle and fat regions of the IR images, resulting in an improved SNR. Another study evaluated six image sets generated from the same raw data: one with FBP and five with IR at various strength levels (1–5) [50]. Muscle attenuation remained stable across all reconstructions, whereas image noise decreased progressively with increasing IR strength. While these two studies reported no significant differences in the attenuation values between FBP- and IR-reconstructed images, the ROI-based approach may have obscured subtle

differences across the entire muscle volume. Moreover, because IR algorithms produce smooth images, particularly at low doses, their impact on BCA metrics warrants further study [47]. Future studies should apply full muscle segmentation to low-dose IR images to clarify these effects.

More recently, deep-learning-based image reconstruction (DLIR) was introduced for low-dose CT, achieving image quality and lesion detectability comparable to those of standard-dose images [51]. However, its specific effect on body-composition metrics has not yet been systematically investigated. Additionally, even with FBP, the choice of the reconstruction kernel can influence the image sharpness and noise. Although soft kernels are typically used in abdominal CT scans to reduce noise, sharp kernels are often used in chest CT scans to better delineate fine pulmonary nodules. Given these factors, the relationship between the reconstruction method, including DLIR and kernel selection, and BCA warrants further exploration.

Slice Thickness

For abdominal CT, a slice thickness of 5 mm or less is generally recommended. Thinner slices improve the spatial resolution and are useful for evaluating fine structures; however, they increase noise and may alter attenuation by reducing voxel averaging. Figure 6 illustrates the BCA results from the same raw data reconstructed at 1-mm and 5-mm slice thickness.

In the phantom study, the SMA remained consistent across thin (1–1.25 mm), medium (2.5–3 mm), and thick (5 mm) sections; however, the SMD increased with thinner slices [41]. Notably, NAMA tended to be smaller, and LAMA tended to be larger in thin-slice images, especially at low tube voltages. A large-scale clinical study of 9,882 CT scans compared 1.25-mm and 5-mm slices and reported strong correlations for the SMD and SMA ($R^2 > 0.97$), with small average differences (-0.1 HU and -0.9 cm², respectively), which were consistent across both nonenhanced and enhanced scans [52]. In contrast, a smaller study of 34 non-contrast PET/CT scans found that the SMA was 1.1%

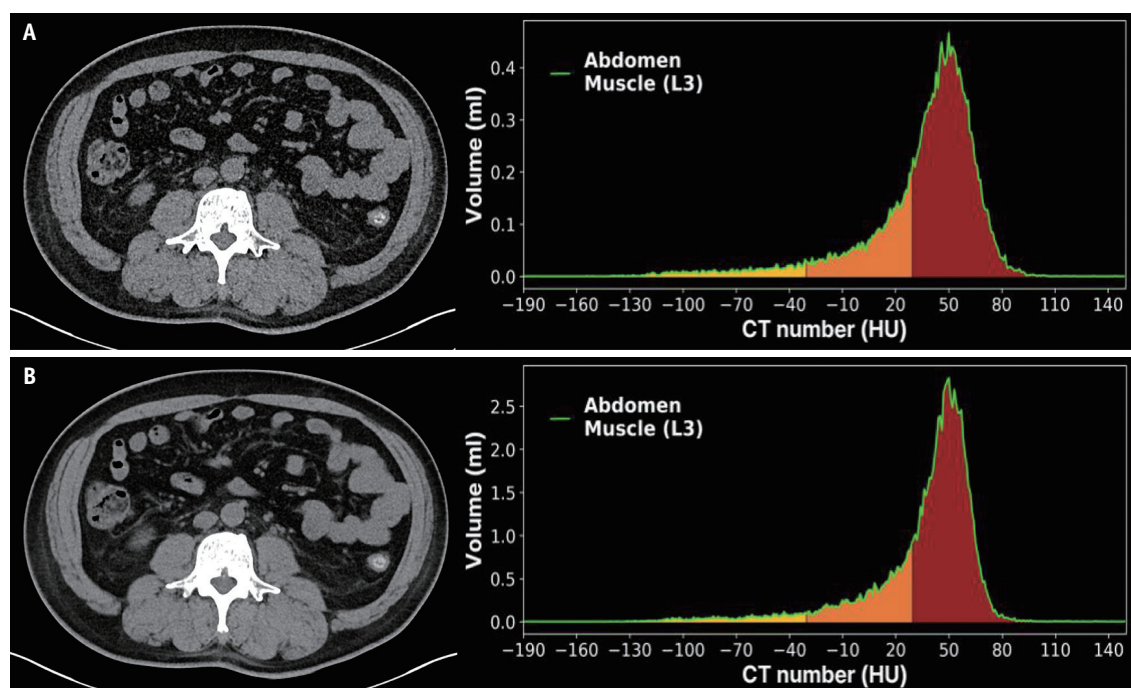


Fig. 6. Portal venous phase abdominal CT in a 50-year-old man, reconstructed from the same acquisition at two slice thicknesses (1 mm vs. 5 mm). **A, B:** The figure shows the BCA results at the L3 level from the CT images reconstructed at 1 mm (**A**) and 5 mm (**B**) from the same raw data. Each panel displays the original CT image (left) and skeletal muscle attenuation histograms (right, NAMA = red; LAMA = orange; IMAT area = yellow). Although slice thickness changes the per-slice volume scale (20% of that on 5-mm images), the pixel-value distribution is broader on the 1-mm images. BCA metrics are reported as 1 mm vs. 5 mm, with areas in cm² and SMD in HU: VAT area 229 vs. 230, SAT area 132 vs. 133, SMA 186 vs. 187, SMD 41.6 ± 22.0 vs. 41.4 ± 21.0 , IMAT area 11 vs. 9, LAMA 44 vs. 41, and NAMA 142 vs. 147. CT = computed tomography, BCA = body-composition analysis, NAMA = normal-attenuation muscle area, LAMA = low-attenuation muscle area, IMAT = intermuscular adipose tissue, SMD = skeletal muscle density, HU = Hounsfield units, VAT = visceral adipose tissue, SAT = subcutaneous adipose tissue, SMA = skeletal muscle area

greater and the SMD was 11.6% lower on 5-mm slices than on 2-mm slices [21].

Another study retrospectively reconstructed the same scan data into various slice thicknesses (2, 3, 4, 5, and 10 mm) and reported a lower SMI in 10-mm images but stable muscle attenuation. However, LAMA was significantly overestimated in thicker slices. In terms of fat, the attenuation and adipose tissue index increased with increasing slice thickness [37]. Similarly, another study found that thinner slices had lower attenuation in all fat compartments; the VAT and IMAT areas were larger in 2-mm slices than in 5-mm slices, whereas the SAT area remained unchanged [42].

While several studies have reported minimal differences between thin and thick slices, others have reported meaningful differences in attenuation and area, especially for LAMA and fat indices. These findings suggest that slice thickness should be standardized or adjusted in BCA studies. The retrospective reconstruction of thin-section images into thicker slices may provide a practical approach for harmonizing results across protocols; however, further validation is required.

TYPES OF CT SCANNERS

DECT

DECT acquires CT data at two energy levels to improve tissue characterization and material differentiation. Various techniques exist, including dual-source DECT, dual-layer detector DECT, and rapid kV-switching systems. By comparing the attenuation at two energies (e.g., low and high kV), DECT enables the differentiation of materials with overlapping CT numbers and allows material-specific reconstruction, such as virtual monoenergetic images and iodine or virtual non-contrast (VNC) images [15,53].

VNC images generated from contrast-enhanced DECT, which subtract the estimated iodine component, are used to reduce radiation exposure by avoiding true non-contrast (TNC) [53]. Given that non-contrast images are preferred in BCA for consistent attenuation-based measurements, several studies have investigated whether VNC images can be substituted for TNC. In a dual-source DECT study, VNC images were generated from the arterial, portal venous, and delayed phases and compared with true TNC images [54]. VNC images from all three phases showed significant differences from TNC images in terms of SMD, SMA, LAMA, and NAMA. Specifically, VNC images tended to have lower SMDs and

higher SMAs than TNC images. In contrast, SAT and VAT attenuations were significantly greater for VNC images than for TNC images, but the areas were either similar or showed only minor differences. Notably, sarcopenia prevalence did not differ between TNC and VNC in the portal venous or delayed phases, suggesting that VNC may be acceptable for sarcopenia classification in certain phases.

However, other studies have reported inconsistent results. One study using rapid kV-switching CT found significantly lower muscle attenuation in VNC images than in TNC images, whereas SAT attenuation remained unchanged [55]. Conversely, a study with dual-source DECT reported no difference in muscle attenuation but greater retroperitoneal fat attenuation in VNC [56]. These discrepancies between the two studies may reflect differences in acquisition parameters, such as the voltage pairing (140/80 kV vs. 150/100 kV) and iodine concentration of the contrast agent (370 mg I/mL vs. 400 mg I/mL). However, the tissue attenuation values from VNC images did not appear to fully match those from the images (Fig. 7). Thus, institutions may need to validate the applicability of VNC images to BCA in their own settings before adopting them as replacements for TNC images.

BCA using virtual monoenergetic images reconstructed at various keV levels (e.g., 40, 55, 70, 85, and 100 keV) have also been explored using rapid kV-switching DECT [57]. Among these, 70 keV images were considered equivalent to 120 kV images on single-energy CT. Muscle and fat attenuation increased with increasing keV, and deviations from the values observed at 70 keV were more pronounced at lower keV values. Similarly, SAT, VAT, and muscle areas varied with the energy level, with greater variation at lower keV values, likely owing to the increased photoelectric effect. Therefore, low-keV monoenergetic images (e.g., <70 keV) should be avoided for BCA, particularly when consistency with conventional CT values is desirable.

PCCT

Multienery CT systems, including photon-counting CT (PCCT) systems, provide more precise spectral separation using energy-resolving detectors [58-60]. PCCT employs semiconductor-based detectors that, unlike conventional energy-integrating detectors, directly convert individual X-ray photons into electrical signals, while categorizing them into multiple energy bins. This design enables the simultaneous acquisition of multi-energy spectral data with enhanced energy resolution.

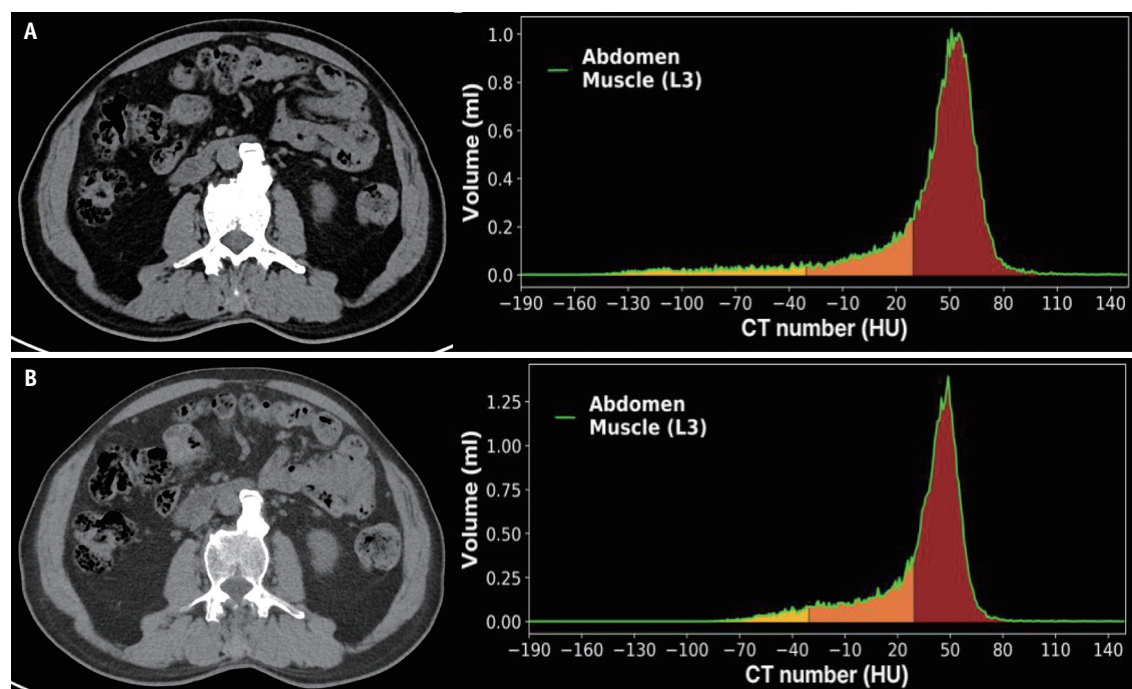


Fig. 7. Abdominal CT in a 75-year-old man acquired on a dual-source dual-energy scanner. **A, B:** The figure shows the BCA results at the L3 level from CT images with TNC images acquired as pre-contrast with single-energy mode (**A**) and VNC images (**B**) generated from portal venous phase images in dual-energy mode (90 and 150 kV). Each panel displays the original CT image (left) and skeletal muscle attenuation histograms (right) (NAMA, red; LAMA, orange; IMAT area, yellow). The VNC histogram showed a slight leftward shift in pixel attenuation relative to the TNC histogram. BCA metrics are reported as TNC vs. VNC, with areas in cm² and SMD in HU: VAT area 154 vs. 126, SAT area 68 vs. 58, SMA 110 vs. 122, SMD 45.7 ± 21.0 vs. 37.4 ± 21.0, IMAT area 8 vs. 6, LAMA 18 vs. 29, and NAMA 92 vs. 93. CT = computed tomography, BCA = body-composition analysis, TNC = true non-contrast, VNC = virtual non-contrast, NAMA = normal-attenuation muscle area, LAMA = low-attenuation muscle area, IMAT = intermuscular adipose tissue, SMD = skeletal muscle density, HU = Hounsfield units, VAT = visceral adipose tissue, SAT = subcutaneous adipose tissue, SMA = skeletal muscle area

Several studies have assessed the reliability of tissue attenuation in VNC images derived from various phases of contrast-enhanced PCCT scans; however, the results for muscles have been inconsistent [61-64]. One study reported higher attenuation in TNC images than in VNC images of the erector spinae, whereas others found lower attenuation, with varying statistical significance. Methodological differences may explain these discrepancies, as some studies have used volumetric segmentation and other ROI sampling methods. Moreover, no study performed full segmentation of the SMA, covering the abdominal wall and paraspinal muscles.

For SAT, two studies reported consistently lower attenuation in TNC images than in VNC images [62,63]. Given that PCCT has only recently been introduced into clinical practice, further research is needed to clarify its role in BCA using the segmentation of muscle and adipose tissue and to standardize quantification methods.

FUTURE DIRECTIONS AND CONCLUSIONS

CT-based BCA, a well-established method for assessing muscle and adipose tissue, may be influenced by technical parameters such as the contrast phase, tube current and voltage, slice thickness, reconstruction algorithm, and scanner type (Table 1). These factors have been reported to affect attenuation- and area-based metrics, potentially limiting the comparability of results across time points, patients, and institutions. Recognizing this parameter-driven variability, mitigation strategies such as standardized acquisition and post-acquisition adjustments to HU could be considered.

At the design level, the variability is most effectively limited in prospective settings using the same scanner and fixed acquisition and reconstruction parameters. In multicenter retrospective studies, such uniformity is rarely feasible. Therefore, explicitly reporting the relevant acquisition and reconstruction settings is essential for characterizing the cohort and supporting reproducibility.

Table 1. Summary of reported effects of CT parameters on BCA metrics

Parameter	SMA	SMD	VAT	SAT	Comments
Contrast enhancement	↑	↑	Area ↓ Attenuation ↑		Effects of contrast agent characteristics and injection protocols remain unexplored
Contrast phase	↑ in the delayed phase	↑ in the delayed phase	↓ in the delayed phase		Gradual SMD increase and VAT area decrease in later phases
Reduced tube current	Mixed (↔ or ↓)	Mixed (↓ or ↑)	Area ↓ Attenuation ↓	Area ↓ Attenuation ↓	Mixed results for SMA and SMD Limited data on the effect of ATCM on BCA
Reduced tube voltage	↑	↑	Area ↑ Attenuation ↓	Area ↑ Attenuation ↓	Effects more pronounced with contrast use
IR vs. FBP	↔	↓	Attenuation ↓	Attenuation ↓	Few studies have used full SMA segmentation IR strength may influence outcomes
Reduced slice thickness	Mixed (↔ or ↓)	↑	↑	↑	Mixed findings likely owing to varied slice thickness

CT = computed tomography, BCA = body-composition analysis, SMA = skeletal muscle area, SMD = skeletal muscle density, VAT = visceral adipose tissue, SAT = subcutaneous adipose tissue, ↑ = increase, ↓ = decrease, Mixed = inconsistent findings across studies, ↔ = no significant change, ATCM = automatic tube current modulation, IR = iterative reconstruction, FBP = filtered back projection

When scanners differ, cross-site calibration using a common phantom can be used to estimate site-specific offsets.

Emerging correction schemes that model kV- and body size-dependent attenuation shifts provide a physics-informed means to align attenuation values across protocols [65]. Likewise, corrections that account for contrast enhancement can be applied when cohorts include both non-contrast and contrast-enhanced scans [23].

Current HU thresholds for muscle and adipose tissue were largely established under 120-kV single-energy conditions; applying them unchanged across heterogeneous protocols may not fully remove bias. Future work should derive thresholds adjusted for protocol differences to match the 120-kV reference. Additionally, because BCA metrics are sensitive to increased noise at lower radiation doses, deep learning-based noise reduction can help stabilize BCA metrics [66]. Deep learning-based image harmonization can also map CT images from different institutions toward a target vendor, dose, or kernel, thereby reducing attenuation differences from different parameters [67]. Finally, generative model-based image conversion may be adapted for abdominal CT to further reduce inconsistencies [68].

In conclusion, CT parameters affect CT-based BCA metrics and are increasingly being recognized as sources of variability. Emphasizing standardized acquisition/reconstruction and using homogeneous parameters when feasible—together with post-acquisition HU adjustments

when heterogeneity is unavoidable—should improve the comparability of CT-based BCA.

Conflicts of Interest

The authors have no potential conflicts of interest to disclose.

Author Contributions

Conceptualization: Moon Hyung Choi. Data curation: all authors. Project administration: Moon Hyung Choi. Supervision: Moon Hyung Choi. Writing—original draft: all authors. Writing—review & editing: all authors.

ORCID IDs

Jin Young Yoo

<https://orcid.org/0000-0003-0007-1960>

Moon Hyung Choi

<https://orcid.org/0000-0001-5962-4772>

Funding Statement

None

REFERENCES

1. Cruz-Jentoft AJ, Bahat G, Bauer J, Boirie Y, Bruyère O, Cederholm T, et al. Sarcopenia: revised European consensus on definition and diagnosis. *Age Ageing* 2019;48:16-31

2. Rosenberg IH. Sarcopenia: origins and clinical relevance. *J Nutr* 1997;127(5 Suppl):990S-991S
3. Liu X, Hao Q, Yue J, Hou L, Xia X, Zhao W, et al. Sarcopenia, obesity and sarcopenia obesity in comparison: prevalence, metabolic profile, and key differences: results from WCHAT study. *J Nutr Health Aging* 2020;24:429-437
4. Donini LM, Busetto L, Bauer JM, Bischoff S, Boirie Y, Cederholm T, et al. Critical appraisal of definitions and diagnostic criteria for sarcopenic obesity based on a systematic review. *Clin Nutr* 2020;39:2368-2388
5. Ceniccola GD, Castro MG, Piovacari SMF, Horie LM, Corrêa FG, Barrere APN, et al. Current technologies in body composition assessment: advantages and disadvantages. *Nutrition* 2019;62:25-31
6. Lee K, Shin Y, Huh J, Sung YS, Lee IS, Yoon KH, et al. Recent issues on body composition imaging for sarcopenia evaluation. *Korean J Radiol* 2019;20:205-217
7. Vogege D, Otto S, Sollmann N, Haggenmüller B, Wolf D, Beer M, et al. Sarcopenia - definition, radiological diagnosis, clinical significance. *Rofo* 2023;195:393-405
8. Chianca V, Albano D, Messina C, Gitto S, Ruffo G, Guarino S, et al. Sarcopenia: imaging assessment and clinical application. *Abdom Radiol (NY)* 2022;47:3205-3216
9. Kim M, Lee SM, Son IT, Park T, Oh BY. Prognostic value of artificial intelligence-driven, computed tomography-based, volumetric assessment of the volume and density of muscle in patients with colon cancer. *Korean J Radiol* 2023;24:849-859
10. Kim DW, Ahn H, Kim KW, Lee SS, Kim HJ, Ko Y, et al. Prognostic value of sarcopenia and myosteatosis in patients with resectable pancreatic ductal adenocarcinoma. *Korean J Radiol* 2022;23:1055-1066
11. Ahn TR, Yoon YC, Kim HS, Kim K, Lee JH. Association between pelvic bone computed tomography-derived body composition and patient outcomes in older adults with proximal femur fracture. *Korean J Radiol* 2023;24:434-443
12. Park HJ, Shin Y, Park J, Kim H, Lee IS, Seo DW, et al. Development and validation of a deep learning system for segmentation of abdominal muscle and fat on computed tomography. *Korean J Radiol* 2020;21:88-100
13. Pickhardt PJ, Summers RM, Garrett JW. Automated CT-based body composition analysis: a golden opportunity. *Korean J Radiol* 2021;22:1934-1937
14. Amini B, Boyle SP, Boutin RD, Lenchik L. Approaches to assessment of muscle mass and myosteatosis on computed tomography: a systematic review. *J Gerontol A Biol Sci Med Sci* 2019;74:1671-1678
15. Tolonen A, Pakarinen T, Sassi A, Kyttä J, Cancino W, Rinta-Kiikka I, et al. Methodology, clinical applications, and future directions of body composition analysis using computed tomography (CT) images: a review. *Eur J Radiol* 2021;145:109943
16. Vasilevska Nikodinovska V, Ivanoski S. Sarcopenia, more than just muscle atrophy: imaging methods for the assessment of muscle quantity and quality. *Rofo* 2023;195:777-789
17. Garcia-Diez AI, Porta-Vilaro M, Isern-Kebschull J, Naude N, Guggenberger R, Brugnara L, et al. Myosteatosis: diagnostic significance and assessment by imaging approaches. *Quant Imaging Med Surg* 2024;14:7937-7957
18. Lortie J, Gage G, Rush B, Heymsfield SB, Szczytutowicz TP, Kuchnia AJ. The effect of computed tomography parameters on sarcopenia and myosteatosis assessment: a scoping review. *J Cachexia Sarcopenia Muscle* 2022;13:2807-2819
19. Mai DVC, Dami I, Pring ET, Gould LE, Lung P, Popuri K, et al. A systematic review of automated segmentation of 3D computed-tomography scans for volumetric body composition analysis. *J Cachexia Sarcopenia Muscle* 2023;14:1973-1986
20. Moeller AR, Garrett JW, Summers RM, Pickhardt PJ. Adjusting for the effect of IV contrast on automated CT body composition measures during the portal venous phase. *Abdom Radiol (NY)* 2024;49:2543-2551
21. Fuchs G, Chretien YR, Mario J, Do S, Eikermann M, Liu B, et al. Quantifying the effect of slice thickness, intravenous contrast and tube current on muscle segmentation: implications for body composition analysis. *Eur Radiol* 2018;28:2455-2463
22. van der Werf A, Dekker IM, Meijerink MR, Wierdsma NJ, de van der Schueren MAE, Langius JAE. Skeletal muscle analyses: agreement between non-contrast and contrast CT scan measurements of skeletal muscle area and mean muscle attenuation. *Clin Physiol Funct Imaging* 2018;38:366-372
23. Perez AA, Pickhardt PJ, Elton DC, Sandfort V, Summers RM. Fully automated CT imaging biomarkers of bone, muscle, and fat: correcting for the effect of intravenous contrast. *Abdom Radiol (NY)* 2021;46:1229-1235
24. McDermott MC, Wildberger JE, Bae KT. Critical but commonly neglected factors that affect contrast medium administration in CT. *Insights Imaging* 2024;15:219
25. Imai K, Ikeda M, Satoh Y, Fujii K, Kawaura C, Nishimoto T, et al. Contrast enhancement efficacy of iodinated contrast media: effect of molecular structure on contrast enhancement. *Eur J Radiol Open* 2018;5:183-188
26. Bae KT. Intravenous contrast medium administration and scan timing at CT: considerations and approaches. *Radiology* 2010;256:32-61
27. Henning MK, Aaløkken TM, Martinsen AC, Johansen S. The impact of body compositions on contrast medium enhancement in chest CT: a randomised controlled trial. *BJR Open* 2023;5:20230054
28. Park HJ, Choi H, Ryu RR. Double low protocol in pediatric abdominal CT for evaluating right lower quadrant pain. *Jpn J Radiol* 2025;43:1176-1185
29. Kim SY, Cho JY, Lee J, Hwang SI, Moon MH, Lee EJ, et al. Low-tube-voltage CT urography using low-concentration-iodine contrast media and iterative reconstruction: a multi-institutional randomized controlled trial for comparison with conventional CT urography. *Korean J Radiol* 2018;19:1119-1129
30. Caruso D. Innovation and optimization of contrast media administration in computed tomography. *Korean J Radiol* 2025;26:210-212

31. Nachit M, Horsmans Y, Summers RM, Leclercq IA, Pickhardt PJ. AI-based CT body composition identifies myosteatosis as key mortality predictor in asymptomatic adults. *Radiology* 2023;307:e222008
32. Rollins KE, Javanmard-Emamghissi H, Awwad A, Macdonald IA, Fearon KCH, Lobo DN. Body composition measurement using computed tomography: does the phase of the scan matter? *Nutrition* 2017;41:37-44
33. Vedder IR, Levolger S, Dierckx RAJO, Viddeleer AR, Bokkers RPH. Effect of contrast phase on quantitative analysis of skeletal muscle and adipose tissue by computed tomography. *Nutrition* 2024;126:112492
34. van Vugt JLA, Coebergh van den Braak RRJ, Schippers HJW, Veen KM, Levolger S, de Bruin RWF, et al. Contrast-enhancement influences skeletal muscle density, but not skeletal muscle mass, measurements on computed tomography. *Clin Nutr* 2018;37:1707-1714
35. Paris MT, Furberg HF, Petruzella S, Akin O, Hötter AM, Mourtzakis M. Influence of contrast administration on computed tomography-based analysis of visceral adipose and skeletal muscle tissue in clear cell renal cell carcinoma. *JPEN J Parenter Enteral Nutr* 2018;42:1148-1155
36. Kim DW, Kim KW, Ko Y, Park T, Lee J, Lee JB, et al. Effects of contrast phases on automated measurements of muscle quantity and quality using CT. *Korean J Radiol* 2021;22:1909-1917
37. Morsbach F, Zhang YH, Martin L, Lindqvist C, Brismar T. Body composition evaluation with computed tomography: contrast media and slice thickness cause methodological errors. *Nutrition* 2019;59:50-55
38. Kaza RK, Platt JF, Goodsitt MM, Al-Hawary MM, Maturen KE, Wasnik AP, et al. Emerging techniques for dose optimization in abdominal CT. *Radiographics* 2014;34:4-17
39. Poletti PA, Platon A, Rutschmann OT, Schmidlin FR, Iselin CE, Becker CD. Low-dose versus standard-dose CT protocol in patients with clinically suspected renal colic. *AJR Am J Roentgenol* 2007;188:927-933
40. Tack D, Sourtzis S, Delpierre I, de Maertelaer V, Gevenois PA. Low-dose unenhanced multidetector CT of patients with suspected renal colic. *AJR Am J Roentgenol* 2003;180:305-311
41. Kim DW, Ha J, Ko Y, Kim KW, Park T, Lee J, et al. Reliability of skeletal muscle area measurement on CT with different parameters: a phantom study. *Korean J Radiol* 2021;22:624-633
42. Troschel AS, Troschel FM, Fuchs G, Marquardt JP, Ackman JB, Yang K, et al. Significance of acquisition parameters for adipose tissue segmentation on CT images. *AJR Am J Roentgenol* 2021;217:177-185
43. Albano D, Camoni L, Rinaldi R, Tucci A, Zilioli VR, Muzi C, et al. Comparison between skeletal muscle and adipose tissue measurements with high-dose CT and low-dose attenuation correction CT of 18F-FDG PET/CT in elderly Hodgkin lymphoma patients: a two-centre validation. *Br J Radiol* 2021;94:20200672
44. Morsbach F, Zhang YH, Nowik P, Martin L, Lindqvist C, Svensson A, et al. Influence of tube potential on CT body composition analysis. *Nutrition* 2018;53:9-13
45. Tong X, Fang X, Wang S, Fan Y, Wei W, Xiao Q, et al. Opportunistic screening for osteoporosis using enhanced images based on dual-energy computed tomography material decomposition: a comparison with quantitative computed tomography. *Quant Imaging Med Surg* 2024;14:352-364
46. Stiller W. Basics of iterative reconstruction methods in computed tomography: a vendor-independent overview. *Eur J Radiol* 2018;109:147-154
47. Patino M, Fuentes JM, Singh S, Hahn PF, Sahani DV. Iterative reconstruction techniques in abdominopelvic CT: technical concepts and clinical implementation. *AJR Am J Roentgenol* 2015;205:W19-W31
48. Kidoh M, Nakaura T, Nakamura S, Oda S, Utsunomiya D, Sakai Y, et al. Low-dose abdominal CT: comparison of low tube voltage with moderate-level iterative reconstruction and standard tube voltage, low tube current with high-level iterative reconstruction. *Clin Radiol* 2013;68:1008-1015
49. Scholtz JE, Wichmann JL, Hüsters K, Beeres M, Nour-Eldin NE, Frellesen C, et al. Automated tube voltage adaptation in combination with advanced modeled iterative reconstruction in thoracoabdominal third-generation 192-slice dual-source computed tomography: effects on image quality and radiation dose. *Acad Radiol* 2015;22:1081-1087
50. Jia Y, Zhai B, He T, Yu Y, Yu N, Duan H, et al. The application of a new model-based iterative reconstruction in low-dose upper abdominal CT. *Acad Radiol* 2019;26:e275-e283
51. Park J, Shin J, Min IK, Bae H, Kim YE, Chung YE. Image quality and lesion detectability of lower-dose abdominopelvic CT obtained using deep learning image reconstruction. *Korean J Radiol* 2022;23:402-412
52. Lee MH, Liu D, Garrett JW, Perez A, Zea R, Summers RM, et al. Comparing fully automated AI body composition measures derived from thin and thick slice CT image data. *Abdom Radiol (NY)* 2024;49:985-996
53. McCollough CH, Leng S, Yu L, Fletcher JG. Dual- and multi-energy CT: principles, technical approaches, and clinical applications. *Radiology* 2015;276:637-653
54. Lee SW, Choi MH, Lee YJ, Choi M. Comparison of muscle and fat measurements between true noncontrast and virtual noncontrast images from three phases of dual-energy dynamic liver CT. *Acad Radiol* 2025;32:5209-5218
55. Borhani AA, Kulzer M, Iranpour N, Ghodadra A, Sparrow M, Furlan A, et al. Comparison of true unenhanced and virtual unenhanced (VUE) attenuation values in abdominopelvic single-source rapid kilovoltage-switching spectral CT. *Abdom Radiol (NY)* 2017;42:710-717
56. Durieux P, Gevenois PA, Muylem AV, Howarth N, Keyzer C. Abdominal attenuation values on virtual and true unenhanced images obtained with third-generation dual-source dual-energy CT. *AJR Am J Roentgenol* 2018;210:1042-1058
57. Toia GV, Garret JW, Rose SD, Szczykutowicz TP, Pickhardt PJ. Comparing fully automated AI body composition biomarkers

- at differing virtual monoenergetic levels using dual-energy CT. *Abdom Radiol (NY)* 2025;50:2758-2769
58. Schwartz FR, Sodickson AD, Pickhardt PJ, Sahani DV, Lev MH, Gupta R. Photon-counting CT: technology, current and potential future clinical applications, and overview of approved systems and those in various stages of research and development. *Radiology* 2025;314:e240662
 59. Greffier J, Villani N, Defez D, Dabli D, Si-Mohamed S. Spectral CT imaging: technical principles of dual-energy CT and multi-energy photon-counting CT. *Diagn Interv Imaging* 2023;104:167-177
 60. Grunz JP, Huflage H. Photon-counting computed tomography: experience in musculoskeletal imaging. *Korean J Radiol* 2024;25:662-672
 61. Dudás I, Schultz L, Benke M, Szűcs Á, Kaposi PN, Szijártó A, et al. The reliability of virtual non-contrast reconstructions of photon-counting detector CT scans in assessing abdominal organs. *BMC Med Imaging* 2024;24:237
 62. Gadsbøll EL, Aurumskjöld ML, Holmquist F, Baubeta E. Virtual non-contrast images in photon-counting computed tomography: impact of different contrast phases. *Acta Radiol* 2024;65:1147-1152
 63. Haag F, Emmrich SS, Hertel A, Rink JS, Nörenberg D, Schoenberg SO, et al. Virtual non-contrast versus true native in photon-counting CT: stability of density of upper abdominal organs and vessels. *Diagnostics (Basel)* 2024;14:1130
 64. Mergen V, Racine D, Jungblut L, Sartoretti T, Bickel S, Monnin P, et al. Virtual noncontrast abdominal imaging with photon-counting detector CT. *Radiology* 2022;305:107-115
 65. Zheng X, Al-Hayek Y, Cummins C, Li X, Nardi L, Albari K, et al. Body size and tube voltage dependent corrections for Hounsfield unit in medical X-ray computed tomography: theory and experiments. *Sci Rep* 2020;10:15696
 66. Tamura A, Mukaida E, Ota Y, Kamata M, Abe S, Yoshioka K. Superior objective and subjective image quality of deep learning reconstruction for low-dose abdominal CT imaging in comparison with model-based iterative reconstruction and filtered back projection. *Br J Radiol* 2021;94:20201357
 67. Lee SB, Hong Y, Cho YJ, Jeong D, Lee J, Choi JW, et al. Enhancing radiomics reproducibility: deep learning-based harmonization of abdominal computed tomography (CT) images. *Bioengineering (Basel)* 2024;11:1212
 68. Hwang HJ, Kim H, Seo JB, Ye JC, Oh G, Lee SM, et al. Generative adversarial network-based image conversion among different computed tomography protocols and vendors: effects on accuracy and variability in quantifying regional disease patterns of interstitial lung disease. *Korean J Radiol* 2023;24:807-820